Dynamic Pricing Optimization for 5G Network Slicing via Three-Part Tariff

Model Specification

Maximize service-provider revenue by learning fixed fee F and per-unit overage fee p of a three-part tariff for URLLC and eMBB slices through reinforcement learning.

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1. Sets and Indices

Symbol	Definition	Range
s	Slice type	$s \in \mathcal{S} = \{U, E\}$ — U: URLLC, E: eMBB
t	Timestep	$t \in \{1, 2, \ldots, T\}$, one step = one hour
i	Individual user	$\mathbb{N}$
$N_{s,t}$	Subscribers in slice s at start of step t	$\mathbb{Z}^+$ (state variable)
$\mathcal{A}_{s,t}$	Active-user set in slice s at step t	set

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2. Three-Part Tariff (3PT) [Ref1, Ref2, Ref3]

Symbol	Index	Range	Role
$F_{s,t}$	s, t	$\mathbb{R}^+$	Fixed fee (learned; action dim 1, 3)
$\bar{Q}_s$	s	$\mathbb{R}^+$	Allowance (pre-fixed constant)
$p_{s,t}$	s, t	$\mathbb{R}^+$	Per-unit overage fee (learned; action dim 2, 4)
$q_{i,s,t}$	i, s, t	$\mathbb{R}^+$	Data usage (random variable)
$B_{i,s,t}$	i, s, t	$\mathbb{R}^+$	Bill (computed)

$$B_{i,s,t} = F_{s,t} + \max\big(0,\ q_{i,s,t} - \bar{Q}_s\big) \cdot p_{s,t} \tag{F1}$$

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3. Data Usage Model [Ref6]

$$q_{i,s,t} \sim \mathrm{LogNormal}(\mu_s,\ \sigma_s^2) \iff \ln q_{i,s,t} \sim \mathcal{N}(\mu_s,\ \sigma_s^2) \tag{F2}$$

$$\mathbb{E}[q] = \exp\!\left(\mu_s + \frac{\sigma_s^2}{2}\right), \qquad \mathrm{Var}[q] = \left[\exp(\sigma_s^2) - 1\right]\ \exp(2\mu_s + \sigma_s^2)$$

Symbol	Index	Range	Meaning
μ_s	s	$\mathbb{R}$	Mean of log-usage
σ_s^2	s	$\mathbb{R}^+$	Variance of log-usage

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4. Timestep Dynamics

Phase A — Agent Decision

Step 1. Observe state.

$$\mathcal{S}_t = \left(N_{U,t},\ N_{E,t},\ \eta_{U,t-1},\ \eta_{E,t-1}\right)$$

Step 2. Select action.

$$a_t = \left(F_{U,t},\ p_{U,t},\ F_{E,t},\ p_{E,t}\right) \in \mathcal{A}$$

Phase B — Subscriber Dynamics

Step 3. Departure. [Ref4, Ref7, Ref8]

$$P_{s,t}^{\mathrm{dep}} = \sigma\!\left(\gamma_{0,s} + \gamma_F\, \tilde{F}_{s,t} + \gamma_p\, \tilde{p}_{s,t} - \gamma_{\eta,s}\, \eta_{s,t-1}\right) \tag{F3}$$

where $\sigma(x) = 1/(1 + e^{-x})$.

Symbol	Index	Sign	Meaning
$\gamma_{0,s}$	s	< 0	Base retention tendency of slice s
γ_F	shared	> 0	Fixed-fee departure sensitivity
γ_p	shared	> 0	Overage-fee departure sensitivity
$\gamma_{\eta,s}$	s	> 0	QoS departure sensitivity; $\gamma_{\eta,U} \gg \gamma_{\eta,E}$
$\tilde{F}_{s,t}$	s, t	—	$F_{s,t}/F_{\text{ref},s}$
$\tilde{p}_{s,t}$	s, t	—	$p_{s,t}/p_{\text{ref},s}$
$\eta_{s,t-1}$	$s, t-1$	—	Previous-step QoS
$F_{\text{ref},s}$	s	$\mathbb{R}^+$	Fixed-fee normalization reference
$p_{\text{ref},s}$	s	$\mathbb{R}^+$	Overage-fee normalization reference

$$\text{Leave}_{i,s,t} \sim \text{Bernoulli}\big(P^{\text{dep}}_{s,t}\big), \quad \forall\, i \in \{1, \ldots, N_{s,t}\}$$

$$N^{\text{surv}}_{s,t} = N_{s,t} - N^{\text{leave}}_{s,t}$$

Step 4. Arrival. [Ref5]

$$P^{\text{arr}}_{s,t} = \sigma\Big(\beta_{0,s} - \beta_F\, \tilde{F}_{s,t} - \beta_p\, \tilde{p}_{s,t}\Big) \tag{F4}$$

Symbol	Index	Sign	Meaning
$\beta_{0,s}$	s	> 0	Base arrival tendency of slice s
β_F	shared	> 0	Fixed-fee arrival sensitivity
β_p	shared	> 0	Overage-fee arrival sensitivity

$$N^{\text{new}}_{s,t} \sim \text{Poisson}\big(\lambda_{\max,s} \cdot P^{\text{arr}}_{s,t}\big) \tag{F5}$$

Symbol	Index	Range	Meaning
$\lambda_{\max,s}$	s	$\mathbb{R}^+$	Max hourly arrival rate

$$\mathcal{A}_{s,t} = \{\text{survivors}\} \cup \{\text{newcomers}\}, \qquad N_{s,t}^{\text{active}} = N_{s,t}^{\text{surv}} + N_{s,t}^{\text{new}} \tag{F6}$$

Phase C — Usage and QoS

Step 5. Usage generation and billing.

$$q_{i,s,t} \sim \text{LogNormal}(\mu_s,\ \sigma_s^2), \qquad B_{i,s,t} = F_{s,t} + \max(0,\ q_{i,s,t} - \bar{Q}_s) \cdot p_{s,t}, \quad \forall\, i \in \mathcal{A}_{s,t}$$

Step 6. QoS realization.

$$\eta_{s,t} \sim \text{Uniform}\big(\eta_s^{\text{low}},\ \eta_s^{\text{high}}\big) \tag{F7}$$

$\eta_{s,(t+1)-1} = \eta_{s,t}$

Symbol	Index	Range	Meaning
$\eta_{s,t}$	s, t	$[0, 1]$	QoS satisfaction
η_s^{tgt}	s	$(0, 1]$	SLA target QoS
η_s^{low}	s	$[0, 1]$	QoS lower bound
η_s^{high}	s	$[0, 1]$	QoS upper bound

Phase D — Learning Signal

Step 7. Reward computation. [Ref9]

$$R_t = \sum_{s \in \{U,E\}} \sum_{i \in \mathcal{A}_{s,t}} B_{i,s,t} \tag{F8}$$

$$\text{Pen}_t = \sum_{s \in \{U,E\}} w_s \cdot N_{s,t}^{\text{active}} \cdot \max\big(0,\ \eta_s^{\text{tgt}} - \eta_{s,t}\big) \tag{F9}$$

Symbol	Index	Range	Meaning
w_s	s	$\mathbb{R}^+$	SLA penalty weight per user
R_t	t	$\mathbb{R}^+$	Total revenue
Pen_t	t	$\mathbb{R}_{\geq 0}$	QoS penalty
r_t	t	$\mathbb{R}$	Reward

$$\boxed{r_t = R_t - \mathrm{Pen}_t} \tag{F10}$$

Step 8. State transition.

$$N_{s,t+1} = N_{s,t}^{\text{active}} \tag{F11}$$

$$\mathcal{S}_{t+1} = \big(N_{U,t+1},\ N_{E,t+1},\ \eta_{U,t},\ \eta_{E,t}\big), \qquad t \leftarrow t+1$$

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5. MDP Definition

Component	Specification
State	$\mathcal{S}_t = (N_{U,t},\ N_{E,t},\ \eta_{U,t-1},\ \eta_{E,t-1})$ — 4-dim
Action	$a_t = (F_{U,t},\ p_{U,t},\ F_{E,t},\ p_{E,t}) \in \mathcal{A}$ — 4-dim
Reward	$r_t = R_t - \mathrm{Pen}_t$
Transition	$N_{s,t+1} = N_{s,t}^{\text{active}}$, QoS carried forward
Initial state	$\mathcal{S}_1 = (N_{U,1},\ N_{E,1},\ \eta_{U,0},\ \eta_{E,0})$
Objective	$\max_\pi\ \mathbb{E}\Big[\sum_{t=1}^{T} \gamma^{t-1} r_t\Big], \quad \gamma \in [0,1)$

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6. Variable Dictionary (41 entries)

Variable	Index	Range	Meaning
s	—	$\{U, E\}$	Slice type
t	—	$\{1, \ldots, T\}$	Timestep
i	—	$\mathbb{N}$	User ID
$N_{s,t}$	s, t	$\mathbb{Z}^+$	Subscribers at start of t
$F_{s,t}$	s, t	$\mathbb{R}^+$	Fixed fee
$p_{s,t}$	s, t	$\mathbb{R}^+$	Overage fee per unit

Variable	Index	Range	Meaning
$\bar{Q}_s$	s	$\mathbb{R}^+$	Allowance (fixed)
$q_{i,s,t}$	i, s, t	$\mathbb{R}^+$	Data usage
$B_{i,s,t}$	i, s, t	$\mathbb{R}^+$	Bill
$\eta_{s,t}$	s, t	$[0, 1]$	QoS satisfaction
$P^{\text{dep}}_{s,t}$	s, t	$[0, 1]$	Departure probability
$P^{\text{arr}}_{s,t}$	s, t	$[0, 1]$	Arrival probability
$N^{\text{surv}}_{s,t}$	s, t	$\mathbb{Z}^+$	Survivors
$N^{\text{new}}_{s,t}$	s, t	$\mathbb{Z}^+$	Newcomers
$N^{\text{active}}_{s,t}$	s, t	$\mathbb{Z}^+$	Active users
$N^{\text{leave}}_{s,t}$	s, t	$\mathbb{Z}^+$	Departed
$\mathcal{A}_{s,t}$	s, t	set	Active-user set
$\gamma_{0,s}$	s	$\mathbb{R}^-$	Base retention tendency
γ_F	shared	$\mathbb{R}^+$	Fixed-fee departure sensitivity
γ_p	shared	$\mathbb{R}^+$	Overage-fee departure sensitivity
$\gamma_{\eta,s}$	s	$\mathbb{R}^+$	QoS departure sensitivity of slice s
$\beta_{0,s}$	s	$\mathbb{R}^+$	Base arrival tendency
β_F	shared	$\mathbb{R}^+$	Fixed-fee arrival sensitivity
β_p	shared	$\mathbb{R}^+$	Overage-fee arrival sensitivity
$\lambda_{\max,s}$	s	$\mathbb{R}^+$	Max hourly arrival rate
μ_s	s	$\mathbb{R}$	Mean of log-usage
σ_s^2	s	$\mathbb{R}^+$	Variance of log-usage
w_s	s	$\mathbb{R}^+$	SLA penalty weight per user
η_s^{tgt}	s	$(0, 1]$	SLA target QoS
η_s^{low}	s	$[0, 1]$	QoS lower bound
η_s^{high}	s	$[0, 1]$	QoS upper bound

Variable	Index	Range	Meaning
$F_{\mathrm{ref},s}$	s	$\mathbb{R}^+$	Fixed-fee normalization reference
$p_{\mathrm{ref},s}$	s	$\mathbb{R}^+$	Overage-fee normalization reference
$\tilde{F}_{s,t}$	s,t	$\mathbb{R}^+$	Normalized fixed fee
$\tilde{p}_{s,t}$	s,t	$\mathbb{R}^+$	Normalized overage fee
R_t	t	$\mathbb{R}^+$	Total revenue
Pen_t	t	$\mathbb{R}_{\geq 0}$	QoS penalty
r_t	t	$\mathbb{R}$	Reward
$\mathcal{S}_t$	t	vector	State
a_t	t	vector	Action
γ	shared	$[0,1)$	Discount factor

7. References

ID	Component	Source
Ref1	3PT structure Eq.(8)–(9)	Wu & Behzad (2023), *Math. Methods Oper. Res.* 97:233–258
Ref2	Zero marginal cost	Wu & Behzad (2023), p.238
Ref3	p_j notation convention	Jiang (2014); pub. Chen, Jiang & Shah (2022), *Int. J. Ind. Organ.* 83:102848
Ref4	Departure vars at $t-1$; churn determinants	Ahn et al. (2006), *Telecom. Policy* 30:552–568
Ref5	Arrival price elasticity sign	Jiang (2014), Table 6; pub. Chen, Jiang & Shah (2022), *Int. J. Ind. Organ.* 83:102848
Ref6	Log-normal traffic	Alasmar et al. (2021), *IEEE/ACM Trans. Netw.* 29(3):1266–1279
Ref7	Logistic regression as churn standard	Ribeiro et al. (2023), *Mgmt Rev. Quarterly* 74:1327–1364
Ref8	URLLC/eMBB QoS requirement gap	3GPP TS 22.261

ID	Component	Source
Ref9	QoS penalty in RL reward	Kim et al. (2019), *Appl. Sci.* 9(11):2361, Eq.(11)